A SIMPLICIAL-NORM LOWER BOUND FOR SOME ONE-RELATOR GROUP

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For a word w in a rank-n free group F_n, we let $\alpha(w)$ denote the class in $H_2(F_n/\langle\langle w\rangle\rangle;\mathbb{Z})$ represented by the presentation complex of $F_n/\langle\langle w\rangle\rangle$. We let $V(w)$ denote the simplicial norm $\|\alpha(w)\|_1$.

Theorem. *For the word*

$$w = \texttt{AABabacbCCBc} \in F(\{a,b,c\}),$$

we have

$$\frac{222}{17} \le V(w^3) \le \frac{670}{51}.$$

It is routine to check that $\mathrm{scl}(w) = 5/4$. Hence, we have

$$\|\alpha_3\|_1 \ge \frac{222}{17} > 13 = 4\,\mathrm{scl}(w^3) - 2,$$

answering a question of Heuer–Löh [1] negatively.

The lower bound comes from an exact parity certificate for every reduced ribbon surface, including signed boundary degrees. The upper bound is obtained by capping a connected threefold cover of an explicitly certified bordered ribbon.

A *ribbon surface* is a compact oriented thickening of a finite labelled graph, mapped to the rose for F_n; we consider $n = 3$ case now. Suppose its boundary components read nonzero powers $w^{k_1}, \dots, w^{k_b}$, with signs allowed. Put

$$\lambda = -\chi(S), \qquad D = \sum_i k_i, \qquad T = \sum_i |k_i|.$$

Call the ribbon *reduced* when the two boundary sides of every spine edge occupy distinct positions in the cyclic word w.

Lemma (Parity certificate)**.** *Every reduced ribbon surface as above admits $K(S) \in \mathbb{Z}_{\ge 0}$ such that*

$$2\lambda - 4D - T \ge \frac{K(S)}{17}, \qquad K(S) \ge \#\{i : k_i \text{ is odd}\}. \tag{1}$$

Proof. Index the letters of w cyclically by $0, \dots, 11$. A state is

$$r = (i, \epsilon, j, \eta), \qquad i \ne j, \quad \epsilon, \eta \in \{1, -1\}, \quad w_i^{\epsilon} = (w_j^{\eta})^{-1}.$$

There are 72 states, with flip $r^* = (j, \eta, i, \epsilon)$. If $r = (i, \epsilon, j, \eta)$, a transition $r \to u$ is allowed when the first position and sign of u are $(j + \eta \bmod 12, \eta)$. There are 216 transitions.

The supplied certificate gives a bit $b(r)$ and an integer $A(r)$ for every state, with $A(r^*) = -A(r)$. Define

$$\omega(j,\eta) = \begin{cases} 1, & (j,\eta) = (11,1) \text{ or } (0,-1), \\ 0, & \text{otherwise,} \end{cases}$$

and, for a transition $r \to u$, set

$$k(r,u) = b(u) \oplus b(r^*) \oplus \omega(j,\eta) \in \{0,1\}.$$

The b-terms telescope around each boundary circuit. The remaining ω-sum is its winding degree modulo two, also for negative degree. Thus the sum $K(S)$ of the corner defects $k(r,u)$ is at least the number of odd-degree boundary components.

Give a transition the integer cost

$$c(r,u) = 408 - 34\big(2(\epsilon+\eta)+1\big) - A(r) - 24k(r,u). \tag{2}$$

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The certificate contains an integer distance matrix d. Exact arithmetic checks

$$d(t,t) = 0, \qquad d(r,t) \le c(r,u) + d(u,t), \qquad c(r,u) + d(u,r) \ge 816$$

for every state t and transition $r \to u$. The middle inequalities telescope along a return path, so every nonempty directed closed walk has cost at least 816. This proves the assertion for walks of arbitrary length, rather than only for enumerated short walks.

For a vertex cycle with q half-edges, allocate

$$\lambda_v = \frac{q-2}{2}, \qquad D_v = \frac{1}{24}\sum(\epsilon+\eta), \qquad T_v = \frac{q}{12}, \qquad K_v = \sum k(r,u).$$

Dividing the closed-walk estimate by 408 gives

$$2\lambda_v - 4D_v - T_v - \frac{K_v}{17} - \frac{1}{408}\sum A(r) \ge 0.$$

Summing over vertices gives $\lambda, D, T, K(S)$, while the A-terms cancel in flip pairs. This proves (1). □

We now apply the surface-diagram formula and reduction theorem of Heuer–Löh [1, Definition 6.2 and Propositions 4.2 and 6.5]. Spherical components may first be discarded: they have degree zero in BG_3, and the negative Euler characteristic ignores them. It then suffices to consider a possibly disconnected reduced diagram on a closed oriented surface Σ, whose faces have nonzero signed degrees $q_1, \dots, q_b$ over the relator w^3. Set

$$Q = \sum_i q_i > 0, \qquad U = \sum_i |q_i|.$$

For these reduced diagrams the negative Euler-characteristic term in the formula is $-\chi(\Sigma)$, and the ribbon inequality is additive over components. Removing the face interiors and reversing the orientation gives a reduced ribbon S with

$$D = 3Q, \qquad T = 3U, \qquad -2\chi(\Sigma) = 2\lambda - 2b.$$

Lemma therefore yields

$$\frac{-2\chi(\Sigma)}{Q} \ge 13 + \frac{(U-Q) + 2(U-b) + K(S)/17}{Q}. \tag{3}$$

Because $3q_i$ is odd exactly when q_i is odd, and because for every integer $d \ge 1$

$$2(d-1) + \frac{d \bmod 2}{17} \ge \frac{d}{17},$$

the second inequality in (1), summed with $d = |q_i|$, gives

$$2(U-b) + \frac{K(S)}{17} \ge \frac{U}{17} \ge \frac{Q}{17}.$$

Also $U - Q \ge 0$. Taking the infimum in the surface-diagram formula proves

$$V_3 \ge 13 + \frac{1}{17} = \frac{222}{17}.$$

Notice that this argument includes negative faces and faces of arbitrary nonzero degree.

References

[1] N. Heuer and C. Löh, *Simplicial volume of one-relator groups and stable commutator length*, Algebr. Geom. Topol. **22** (2022), 1615–1661. doi:10.2140/agt.2022.22.1615.

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